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IIT JEE | MEDICAL | FOUNDATION

CODE A

Monthly Online Test - 2 | [JEE 2022]

Date : 29/06/2020 **Maximum Marks: 300**

General Instructions

1. The test is of **3 hours** duration and the maximum marks are **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **Five Integer value type questions.** Each question has an integer value ranging from **0 – 999.**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Syllabus:

Physics : Introduction to Vector & Forces and Kinematics of a Particle.

Chemistry : Stoichiometry and Atomic Structure.

Mathematics : Quadratic Equation and Sequence and Series.

SUBJECT I: PHYSICS	Suggested Time: 60 Minutes	100 MARKS

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The torque of force $\vec{F} = (2\hat{i} - 3\hat{j} + 4\hat{k})$ newton acting at the point $\vec{r} = (3\hat{i} + 2\hat{j} + 3\hat{k})$ metre about origin is: (in N-m)

(A) $6\hat{i} - 6\hat{j} + 12\hat{k}$ **(B)** $17\hat{i} - 6\hat{j} - 13\hat{k}$ **(C)** $-6\hat{i} + 6\hat{j} - 12\hat{k}$ **(D)** $-17\hat{i} + 6\hat{j} + 13\hat{k}$

2. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A} + \vec{B} = \vec{C}$ and $A^2 + B^2 = C^2$. If θ is the angle between positive direction of $\vec{A}$ and $\vec{B}$ then the correct statement is:

(A) $\theta = \pi$ **(B)** $\theta = \frac{2\pi}{3}$ **(C)** $\theta = 0$ **(D)** $\theta = \frac{\pi}{2}$

3. If the angle between the vectors $\vec{A}$ and $\vec{B}$ is θ, the value of the product $(\vec{B} \times \vec{A}) \cdot \vec{A}$ is equal to:

(A) $BA^2 \cos\theta$ **(B)** $BA^2 \sin\theta$ **(C)** $BA^2 \sin\theta \cos\theta$ **(D)** zero

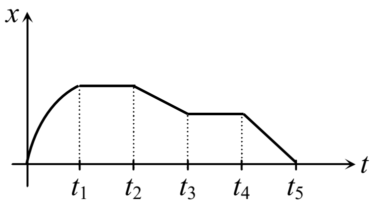
4. A ball is thrown vertically upwards from the ground and a student gazing out of the window sees it moving upward past him at $10\ \text{ms}^{-1}$. The window is at 15 m above the ground level. The velocity of ball 3 s after it was projected from the ground is: [Take $g = 10\ \text{ms}^{-2}$]

(A) 10 m/s, up **(B)** $20\ \text{ms}^{-1}$, up **(C)** $20\ \text{ms}^{-1}$, down **(D)** $10\ \text{ms}^{-1}$, down

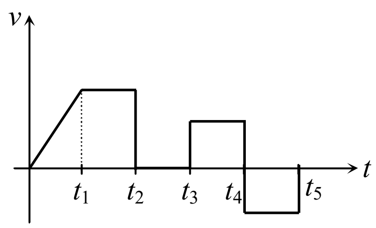
5. A river is flowing from west to east with a speed of 5m min^{-1}. A man can swim in still water with a velocity $10\ \text{m min}^{-1}$. In which direction should the man swim so as to take the shortest possible path to go to the south?

(A) 30° east of south **(B)** 30° west of south
(C) south **(D)** 30° west of north

6. For position-time(x-t) curve as shown in figure, the velocity-time (v-t) curve will be (from 0 to t_1 curve is parabolic)



(A)



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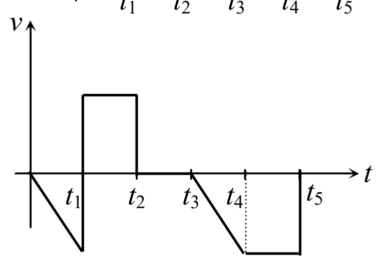
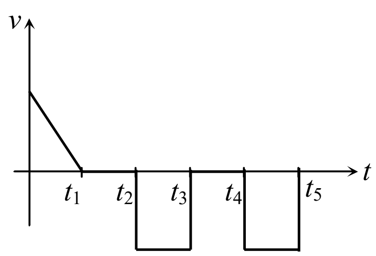
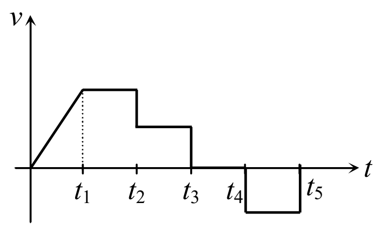


(C)



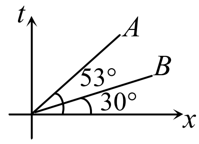
(D)



7. A particle located at $x = 0$ at time $t = 0$, starts moving along the positive x-direction with a velocity 'v' that varies as $v = \alpha\sqrt{x}$. The displacement of the particle varies with time as.

(A) t^3 **(B)** t^2 **(C)** t **(D)** $t^{1/2}$

8. Time - displacement (t-x) graph of two objects A and B is shown in figure. The ratio of their speeds (v_A / v_B) is: $(\tan 37° = 3/4)$



(A) $\frac{\sqrt{3}}{4}$ **(B)** $\frac{4}{\sqrt{3}}$ **(C)** $\frac{4}{3\sqrt{3}}$ **(D)** $\frac{3\sqrt{3}}{4}$

9. Rain is falling vertically downwards with a velocity of 3 km/hr. A man walks in the rain with a velocity of 4 km/hr. The raindrops will fall on the man with a velocity of :

(A) 1 km/hr **(B)** 3 km/hr **(C)** 4 km/hr **(D)** 5 km/hr

10. The acceleration of a particle, starting from rest, varies with time according to the relation $a = kt + c$. The velocity v of the particle after a time t will be:

(A) $kt^2 + ct$ **(B)** $\frac{1}{2}(kt^2 + ct)$ **(C)** $\frac{1}{2}(kt^2 + 2ct)$ **(D)** $kt^2 + \frac{1}{2}ct$

11. A body is thrown up in a lift with an upward velocity u relative to the lift from its floor and the time of flight is found to be t. The acceleration of the lift will be:

(A) $\frac{u - gt}{2}$ **(B)** $\frac{u + gt}{2}$ **(C)** $\frac{2u - gt}{t}$ **(D)** $\frac{u}{t} - g$

12. A man can row a boat with speed 4 km/hr in still water. If the velocity of water in river is 3 km/hr. The time taken to reach just opposite end is: (river width = 500 m)

(A) $\frac{500}{\sqrt{7}}$ hr **(B)** $\frac{1}{2\sqrt{7}}$ hr **(C)** 100 hr **(D)** None of these

13. Two bodies are projected vertically upwards from one point with the same initial velocities v_0 m/s. The second body is thrown τ s after the first. The two bodies meet after time:

(A) $\frac{v_0}{g} - \frac{\tau}{2}$ **(B)** $\frac{v_0}{g} + \tau$ **(C)** $\frac{v_0}{g} + \frac{\tau}{2}$ **(D)** $\frac{v_0}{2g} + \tau$

14. A man running uniformly at 8 m/s is 16 m behind a bus when it starts accelerating at 2 ms^{-2}. Time taken by him to board the bus is:

(A) 2s **(B)** 3s **(C)** 4s **(D)** 5s

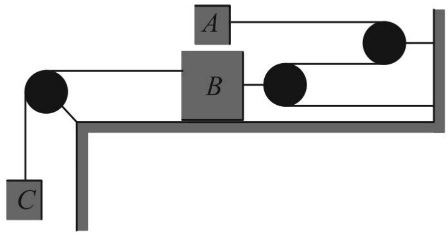
15. The force required to just move a body up the inclined plane is double the force required to just prevent the body from sliding down the plane. The coefficient of friction is μ. If θ is the angle of inclination of the plane than $\tan\theta$ is equal to:

(A) μ **(B)** 3μ **(C)** 2μ **(D)** 0.5μ

16. A particle is placed at rest inside a hollow hemisphere of radius R. The coefficient of friction between the particle and the hemisphere is $\mu = \frac{1}{\sqrt{3}}$. The maximum height up to which the particle can remain stationary is:

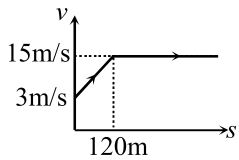
(A) $\frac{R}{2}$ **(B)** $\left(1 - \frac{\sqrt{3}}{2}\right)R$ **(C)** $\frac{\sqrt{3}}{2}R$ **(D)** $\frac{3R}{8}$

17. What is the largest mass of C in kg that can be suspended without moving blocks A and B? The static coefficient of friction for all plane surface of contact is 0.3. Mass of block A is 50 kg and block B is 70kg. Neglect friction in the pulleys:



(A) 120 kg **(B)** 92 kg **(C)** 81 kg **(D)** None of these

18. The velocity-displacement graph of a particle is as shown in figure. The acceleration of the particle when displacement is 100 m will be:



(A) 1.3 m/s^2 **(B)** 1 m/s^2 **(C)** 1 m/s^2 **(D)** 0.13 m/s^2

19. Unit vector parallel to the resultant of vectors $\vec{A} = 4\hat{i} - 3\hat{j}$ and $\vec{B} = 8\hat{i} + 8\hat{j}$ will be :

(A) $\frac{24\hat{i}+5\hat{j}}{13}$ **(B)** $\frac{12\hat{i}+5\hat{j}}{13}$ **(C)** $\frac{6\hat{i}+5\hat{j}}{13}$ **(D)** None of these

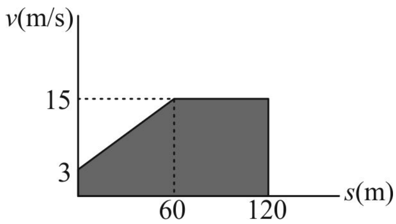
20. If a vector $2\hat{i}+3\hat{j}+8\hat{k}$ is perpendicular to the vector $4\hat{j}-4\hat{i}+\alpha\hat{k}$, then the value of α is:

(A) -1 **(B)** $\frac{1}{2}$ **(C)** $-\frac{1}{2}$ **(D)** 1

SECTION 2

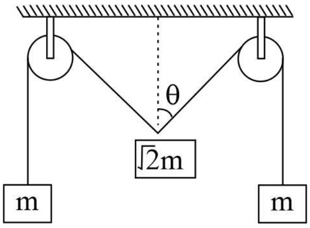
This section contains 5 Integer value type questions. Each question has an integer value ranging from 0 – 999.

21. The v-s graph describing the motion of a motorcycle is shown in figure. Determine the time $(in\,seconds)$ needed for the motorcycle to reach the position $s = 120\text{m}$. Given $\ln 5 = 1.6$.

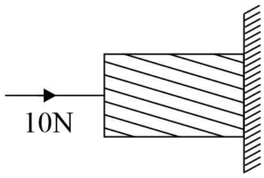


22. Two identical balls are shot upward one after another at an interval of 2s along the same vertical line with same initial velocity of 40ms^{-1}. The height at which the balls collide is: (in metres)

23. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium the angle θ should be $\frac{n\pi}{4}$ radian. Find n.



24. A horizontal force of 10N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is: (in Newton)



25. Maximum and minimum values of the resultant of two forces acting at a point are 7 N and 3 N respectively. The smaller force will be equal to: (in N)

SUBJECT II: CHEMISTRY	Suggested Time: 45 Minutes	100 MARKS

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Calcium combines with oxygen to form calcium oxide. How many litres of oxygen (at STP) are required to make $112\,g$ of CaO ? (M_0 of $Ca = 40\,g, O = 16\,g$)

(A) 22.4 L **(B)** 44.8 L **(C)** 11.2 L **(D)** 112 L

2. 500 mL of 0.1 N of $AgNO_3$ is added to 500 mL of 0.1 N KCl solution.

What is the concentration of nitrate ions in the resulting solution assuming no change in volume ?

(A) 0 N **(B)** 0.1 N **(C)** 0.2 N **(D)** 0.05 N

3. Find the amount of 98% pure Na_2CO_3 required to prepare 5L of 1 N solution.
(M_0 of $Na_2CO_3 = 106\,g$)

(A) 265 g **(B)** 270.41 g **(C)** 5 g **(D)** 135.2 g

4. You are given $1L$ of $2\,M\,HCl$ and $1L$ of $5\,M\,HCl$ solutions. What is the maximum volume of $3\,M\,HCl$ which you can make from the above solutions (without addition of water) ?

(A) $2L$ **(B)** $1.5L$ **(C)** $3L$ **(D)** $4L$

5. A mixture of KOH and Na_2CO_3 required 20 mL of $\frac{N}{20}HCl$ solution when titrated using phenolphthalein as indicator. But the same amount of solution required 30 mL of the same acid solution when titrated using methyl orange indicator. Calculate the amount of KOH in the mixture.
(M_0 of $KOH = 56\,g$ & $Na_2CO_3 = 106\,g$)

(A) $0.106\,g$ **(B)** $0.5\,g$ **(C)** $0.53\,g$ **(D)** $0.028\,g$

6. When 400 g of a 20% (w/w) solution was cooled, 50 g of solute precipitated. What is the (w/w) percentage concentration of remaining solution?

(A) 7.5% **(B)** 8.57%
(C) 2.5% **(D)** Cannot be determined

7. What volume of hydrogen gas (at STP) will be liberated by the reaction of excess Zn on 50 mL dilute H_2SO_4 of specific gravity = 2 (g/ml) and w/w percentage 49% ?

(A) 11200 L **(B)** 11.2 L **(C)** 22.4 L **(D)** 44.8 L

8. A 100 mL solution of 0.1 M of a metal chloride MCl_x requires 50 mL of 0.6 M $AgNO_3$ solution for complete precipitation of AgCl. The value of x is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

9. One mol potassium chlorate is thermally decomposed to produce $KCl(s)$ and $O_2(g)$ and excess of aluminium is burnt in the gaseous product. How many moles of aluminium oxide are formed?

(A) 1 **(B)** 2 **(C)** 1.5 **(D)** 3

10. A transition metal complex shows a spin only magnetic moment of 3.87 BM at room temperature. The number of unpaired electrons in the metal is:

(A) 3 **(B)** 4 **(C)** 5 **(D)** 2

11. λ_0 is the threshold wavelength of a metal for photoelectric emission. If the metal is exposed to light of wavelength λ, then the velocity of ejected electron will be $\sqrt{\frac{2hk}{m}(\lambda_0 - \lambda)}$. Value of k is :

(A) speed of light **(B)** 1 **(C)** $\frac{c}{\lambda_0 \lambda}$ **(D)** $\frac{1}{\lambda \lambda_0}$

12. The radial probability distribution function curve obtained for an orbital wave function (ψ) has 3 peaks and 2 radial nodes. The valence electron of which one of the following metals does this wave function (ψ) correspond to?

(A) Ca **(B)** Li **(C)** K **(D)** Na

13. The wave number corresponding to highest energy transition in Balmer series, in the emission spectra of hydrogen represented by: $(R_H = 109737\,cm^{-1})$

(A) $4389.48\,cm^{-1}$ **(B)** $2194.74\,cm^{-1}$ **(C)** $5486.85\,cm^{-1}$ **(D)** $27434.25\,cm^{-1}$

14. How many electrons are there in the M-shell of an element with atomic number, $Z = 24$?

(A) 5 **(B)** 6 **(C)** 12 **(D)** 13

15. Which of the following is not possible for 4p electrons?

I. $n = 4 \ , \ \ell = 1 \ , \ m = +1 \ , \ s = -\frac{1}{2}$

II. $n = 4 \ , \ \ell = 1 \ , \ m = 0 \ , \ s = +\frac{1}{2}$

III. $n = 4 \ , \ \ell = 0 \ , \ m = 0 \ , \ s = +\frac{1}{2}$

IV. $n = 4 \ , \ \ell = 1 \ , \ m = +2 \ , \ s = -\frac{1}{2}$

(A) Only I and II **(B)** Only II and III **(C)** Only III **(D)** Only III and IV

16. The ratio of ionisation energy of H and Be^{3+} is : (Atomic number of H = 1, Be = 4)

(A) 1 : 4 **(B)** 1 : 3 **(C)** 1 : 9 **(D)** 1 : 16

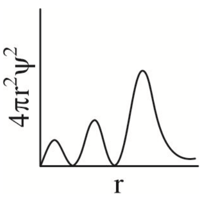
17. The quantum number 'm' of a free gaseous atom is associated with :

(A) The size of the orbital
(B) The shape of the orbital
(C) The spatial orientation of the orbital
(D) The energy of the orbital in the absence of magnetic field

18. The de-Broglie wavelength (in m) of a tennis ball of mass 120g moving with a velocity of $20\,m/s$ is approximately. $(h = 6.6 \times 10^{-34}\,Js)$

(A) 2.75×10^{-37} **(B)** 2.75×10^{-34} **(C)** 2.75×10^{-31} **(D)** 1.69×10^{-40}

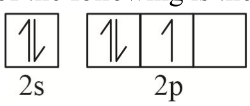
19. The following graph is plotted for ns-orbital.

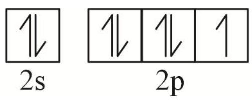


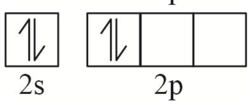
The value of n will be:

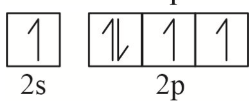
(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

20. Which of the following is the correct arrangement of electrons ?

(A) 

(B) 

(C) 

(D) 

SECTION 2

This section contains 5 Integer value type questions. Each question has an integer value ranging from 0 – 999.

21. $1.2475\,g$ of $CuSO_4 \cdot x\,H_2O$ was dissolved in water and H_2S was passed till CuS was completely precipitated. The filtrate contained the liberated H_2SO_4 which required 20 mL of $\frac{N}{2}$NaOH for complete neutralisation. What is the value of 'x'? (M_0 of $CuSO_4 = 159.5$)

22. 'x'g of sulphur has as many atoms as in 3g of carbon. The value of x is:
(M_0 of $C = 12$, M_0 of $S = 32$)

23. Sodium chloride of 70.2% purity is used to produce salt cake, Na_2SO_4 of 85.2% purity according to the equation $2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$.
Calculate the number of kg of NaCl (impure) required to produce 1 kg of salt cake (impure).
(M_0 of $Na = 23\,g$, $S = 32\,g$, $O = 16\,g$, $Cl = 35.5\,g$)

24. What will be the number of angular nodes for 4f atomic orbital ?

25. The decomposition of a certain mass of $CaCO_3$ gave $11.2\,dm^3$ of CO_2 gas at STP. The mass of KOH in g required to completely neutralize the gas according to reaction $2KOH + CO_2 \longrightarrow K_2CO_3 + H_2O$ is:
(M_0 of $CaCO_3 = 100\,g$, $KOH = 56\,g$)

SUBJECT III: MATHEMATICS	Suggested Time: 75 Minutes	100 MARKS

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Let α, β are roots of $x^2 - x - 1 = 0$ Then $\alpha^5 + \beta^5$ is:

(A) 7 **(B)** 11 **(C)** 8 **(D)** 1

2. If the sum of an Infinite G.P. is x. Whose terms are decreasing and first then is 5, then x Belongs to:

(A) $(5, \infty)$ **(B)** $\left(\frac{5}{2}, \infty\right) - \{5\}$ **(C)** $\left(\frac{-5}{2}, \frac{5}{2}\right)$ **(D)** $\left(0, \frac{5}{2}\right)$

3. The greatest integral value of p so that 6 lies between the roots of equation $x^2 + 2(p-3)x + 9 = 0$ is:

(A) -1 **(B)** 0 **(C)** 2 **(D)** 3

4. If A$\in$I and $(x-A)(x-12)+2=0$ has integral roots then the sum of all possible values of A is:

(A) 15 **(B)** 6 **(C)** 9 **(D)** 24

5. If the 9^{th} term of an A.P. is zero, then the ratio of its 29^{th} term to 19^{th} term is:

(A) $1:2$ **(B)** $2:1$ **(C)** $1:3$ **(D)** $3:1$

6. If $x, y \in R, x > y$ and $xy = 1$ Then minimum possible value of $\frac{x^2 + y^2}{x - y}$ is:

(A) $4\sqrt{2}$ **(B)** $8\sqrt{2}$ **(C)** $2\sqrt{2}$ **(D)** $\sqrt{2}$

7. The corresponding terms of two A.P.s are multiplied together to give series 240, 336, 440, …. Find 10^{th} term of series:

(A) 1392 **(B)** 1293 **(C)** 1239 **(D)** 3912

8. If minimum$\left(2x^2 - Ax + 2\right) >$ maximum$\left(B - 1 + 2x - x^2\right) A, B \in R,$ then the roots of equation $2x^2 + Ax + (2 - B) = 0$ are:

(A) Positive and distinct **(B)** Negative and distinct
(C) Opposite in sign **(D)** complex

9. For A, B, C$\in$R and $B^2 \geq 4AC$, if all roots of $Ax^4 + Bx^2 + C = 0$ are real then:

(A) $A < 0, B > 0, C > 0$ **(B)** $A > 0, B < 0, C > 0$
(C) $A > 0, B > 0, C > 0$ **(D)** $A < 0, B < 0, C > 0$

10. There are five numbers in which first three numbers are in A.P. and three numbers in middle are in G.P., If number in odd places are in G.P., then number in last three places are in:

(A) $A \cdot P \cdot$ **(B)** $G \cdot P \cdot$ **(C)** $H \cdot P \cdot$ **(D)** $N \cdot O \cdot T \cdot$

11. If each side of a square sheet Be "a" and four corners are cut away equally to form a regular octagon having each side "x" then $\sum_{K=0}^{\infty} \left(\frac{x}{a}\right)^K$ is:

(A) $2 - \sqrt{2}$ **(B)** $2 + \sqrt{2}$ **(C)** $\frac{1}{2 - \sqrt{2}}$ **(D)** $\frac{1}{2 + \sqrt{2}}$

12. If equation $x^2 - Px + Q = 0$ and $x^2 - Ax + B = 0$ have a common root and the other root of second equation is reciprocal of other root of the first equation then $(Q - B)^2$ is:

(A) $AQ(P - B)^2$ **(B)** $BQ(P - A)^2$ **(C)** $BQ(P - B)^2$ **(D)** $ABQ(P - B)^2$

13. Let $a_1, a_2, a_3, \ldots a_{n-1}, a_n$ be in $G \cdot P \cdot$ Such that $3a_1 + 7a_2 + 3a_3 - 4a_5 = 0$, then common ratio of $G \cdot P \cdot$ can be:

(A) 2 **(B)** $\frac{3}{2}$ **(C)** $\frac{5}{2}$ **(D)** $\frac{1}{2}$

14. Let $A_1, A_2, A_3, \ldots$ be in $A \cdot P \cdot$ and $G_1, G_2, G_3, \ldots Be$ in $G \cdot P \cdot$. If $A_1 = G_1 = 2$ and $A_{10} = G_{10} = 3$ then:

(A) $A_7 G_{19} > A_{19} G_{10}$ **(B)** $A_7 G_{19} < A_{19} G_{10}$

(C) $A_7 G_{19} + A_{19} G_{10} = 24$ **(D)** $A_7 G_{19} + A_{19} G_{10} = 18$

15. If α, β, γ are roots of equation $4x^3 + 5x^2 + 7x + 3 = 0$ then the value of

$\frac{1}{(1-\alpha)(1-\beta)} + \frac{1}{(1-\beta)(1-\gamma)} + \frac{1}{(1-\gamma)(1-\alpha)}$ is:

(A) $\frac{17}{13}$ **(B)** $\frac{7}{9}$ **(C)** $\frac{17}{19}$ **(D)** $\frac{7}{13}$

16. If equation $4x^2 - 8x + 3 = 0, x^2 + Ax - 1 = 0$ and $2x^2 + x + B = 0$ may have a common root for each pair of equation and all the equations have not a single common Root, then $12A + B$ is:

(A) 17 **(B)** 12 **(C)** –11 **(D)** –16

17. If $A \in (0,1]$ and satisfy $A^{2008} - 2A + 1 = 0$ and $s = 1 + A + A^2 + \ldots + A^{2006} + A^{2007}$, then find sum of all possible values of s :

(A) 2008 **(B)** 2009 **(C)** 2010 **(D)** 2

18. If e^t and e^{-t} are roots of $3x^2 - (A+B)x + 2A = 0$ Given $A, B, t \in R, t \neq 0$ then least integral value of B is:

(A) 4 **(B)** 5 **(C)** 9 **(D)** 10

19. If natural numbers are written as a sequence of digits 123456789101112… then 2007$^{\text{th}}$ digits is:

(A) 0 **(B)** 7 **(C)** 4 **(D)** 5

20. Numbers of real solution of equation: $\sqrt{2x^2 + 3x + 5} + \sqrt{2x^2 - 3x + 5} = 3x$ is:

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

SECTION 2

This section contains 5 Integer value type questions. Each question has an integer value ranging from 0 – 999.

21. $s = \sum_{n=2}^{\infty} \frac{3n^2 + 1}{(n^2 - 1)^3} = \frac{p}{q}$ Where p, q are co-prime Find $p + q$.

22. Let $0 < x < y < 125$, Find number of pairs of Integers (x, y) Such that $A \cdot M \cdot$ of x, y exceeds their $G \cdot M \cdot$ By 2.

23. The Quadratic polynomial $f(x)$ satisfy equation $f(x) - f(x+1) = 2x - 1 \quad \forall x \in R$. If the Greatest value of $f(x)$ in $(-\infty, 0]$ is 2, then find $f(-1) + f(1)$.

24. Let $A < B < C$ And C is multiple of B. If A, B, C in $H \cdot P \cdot$ and $A = 20$ then number of values of B is:

25. Let A and B be arbitrary real numbers, Find the smallest Natural number B for which the equation $x^2 + 2(A+B)x + (A - B + 8) = 0$ has unequal real roots for all $A \in R$.

End of MOT – 2 | JEE 2022